

TC-Net: A joint learning framework based on CNN and vision transformer for multi-lesion medical images segmentation

Zhongxiang Zhang^a, Guangmin Sun^a, Kun Zheng^a, Jin-Kui Yang^{b,c}, Xiao-rong Zhu^{b,c}, Yu Li^{a,d,*}

^a The Faculty of Information Technology, Beijing University of Technology, China

^b The Department of Endocrinology, Beijing Tongren Hospital, China

^c Capital Medical University, China

^d The Beijing Institute of Artificial Intelligence, Beijing University of Technology, China

ARTICLE INFO

Keywords:

Medical image segmentation
Convolutional neural network
Vision transformer
Class-imbalance

ABSTRACT

Background: With the rapid advancement of medical imaging technology, the demand for accurate segmentation of medical images is increasing. However, most existing methods are unable to capture locality and long-range dependency information in integrated ways for medical images.

Method: In this paper, we propose an elegant segmentation framework for medical images named TC-Net, which can utilize both the locality-aware and long-range dependencies in the medical images. As for the locality-aware perspective, we employ a CNN-based encoder and decoder structure. The CNN branch uses the locality of convolution operations to dig out local information in medical images. As for the long-range dependencies, we construct a Transformer branch to focus on the global context. Additionally, we proposed a locality-aware and long-range dependency concatenation strategy (LLCS) to aggregate the feature maps obtained from the two subbranches. Finally, we present a dynamic cyclical focal loss (DCFL) to address the class imbalance problem in multi-lesion segmentation.

Results: Comprehensive experiments were conducted on lesion segmentation tasks using two fundus image databases and a skin image database. The TC-Net achieves scores of 0.6985 and 0.5171 in the metric of mean pixel accuracy on the IDRiD and DDR databases, respectively. Moreover, on the skin image database, the TC-Net reached mean pixel accuracy of 0.8886. The experiment results demonstrate that the proposed method achieves better performance than other deep learning segmentation schemes. Furthermore, the proposed DCFL achieves higher performance than other loss functions in multi-lesion segmentation.

Significance: The proposed TC-Net is a promising new framework for multi-lesion medical image segmentation and many other challenging image segmentation tasks. © 2001 Elsevier Science. All rights reserved.

1. Introduction

Accurate multi-lesion and multi-organ segmentation play an essential role in disease analysis and clinical surgery [1]. Recently, the development of accurate and efficient segmentation algorithms for medical images has become a popular focus of research, and an abundance of algorithms have emerged for various types of medical image segmentation in the past few years, such as brain segmentation [2,3], lung segmentation [4], cell segmentation [5] and liver segmentation [6,7]. Recently, the development of accurate and efficient segmentation algorithms for medical images has become a popular focus of research.

Recent medical image segmentation approaches mainly depend on

convolutional neural networks (CNNs) or Vision Transformers. The typical CNN-based methods rely on networks with encoder-decoder structures, such as U-Net [8], SegNet [9], KiU-Net [10] and 3D U-Net [11]. The outstanding performance of these CNN-based methods proves that CNNs have a powerful ability to learn discriminative representations for medical image segmentation. Inspired by the Transformer model's remarkable performance in natural language processing (NLP), researchers have also recently attempted to introduce Transformers into the field of computer vision. In particular, several Transformer-based segmentation approaches have been proposed for medical image segmentation. Networks such as Swin-UNet [12], TransUNet [13] and Multi-Compound Transformer (MCTrans) [14] have been proposed to

* Corresponding author. The Faculty of Information Technology, Beijing University of Technology, China.

E-mail addresses: zxx@emails.bjut.edu.cn (Z. Zhang), gmsun@bjut.edu.cn (G. Sun), zhengkun@bjut.edu.cn (K. Zheng), jinkui.yang@gmail.com (J.-K. Yang), zhuxr_06@sina.com (X.-r. Zhu), yuli@bjut.edu.cn (Y. Li).

<https://doi.org/10.1016/j.combiomed.2023.106967>

Received 30 December 2022; Received in revised form 12 April 2023; Accepted 19 April 2023

Available online 13 May 2023

0010-4825/© 2023 Elsevier Ltd. All rights reserved.

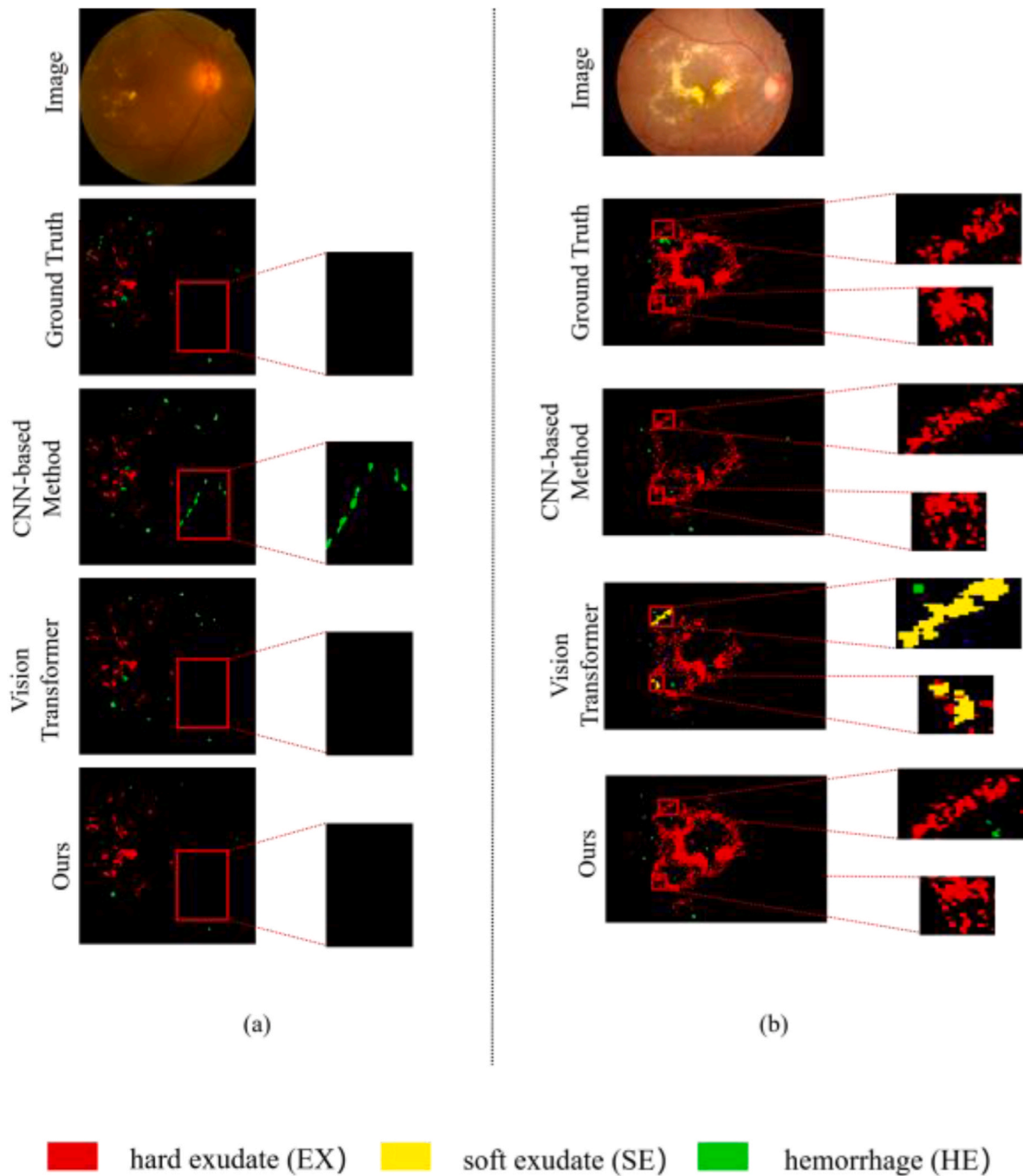


Fig. 1. Example of multi-lesion segmentation results obtained with a CNN-based method, a Vision Transformer and our method.

analyze various medical image modalities. These methods employ Transformers to encode long-range dependencies and learn representations that capture highly expressive information.

Despite the considerable advances achieved in medical image segmentation, however, most existing methods cannot take full advantage of the rich locality and long-range dependency information contained in medical images. Due to the intrinsically local nature of the convolution operation, it is difficult for CNN-based segmentation methods to model the global and long-range interactions of semantic information [15]. Unlike CNN, Transformers utilize dispense convolution operators and attention mechanisms which can model global contexts. The experiment shows that long-range dependency is crucial for medical image segmentation, which can effectively avoid the misclassification of lesions and background pixels. The case of multi-lesion segmentation in fundus

images is an example that illustrates the deficiencies of the existing medical image segmentation methods.

As shown in Fig. 1 (a), the receptive fields of CNN-based methods are not adequate to capture the global features of medical images; consequently, such a method mistakenly divides a large background area into hemorrhage (HE). However, Transformers map a series of image patch sequences into abstract continuous representations, which obtain global dependency on the entire image. As shown in Fig. 1 (a), Transformer effectively avoids the wrong segmentation results of the background in fundus images. Therefore, it is desirable to capture the long-range dependencies in medical images.

Recently, some researchers have extended Transformers into the computer vision domain to learn long-range dependencies. However, compared with CNN-based approaches, Transformers lack the

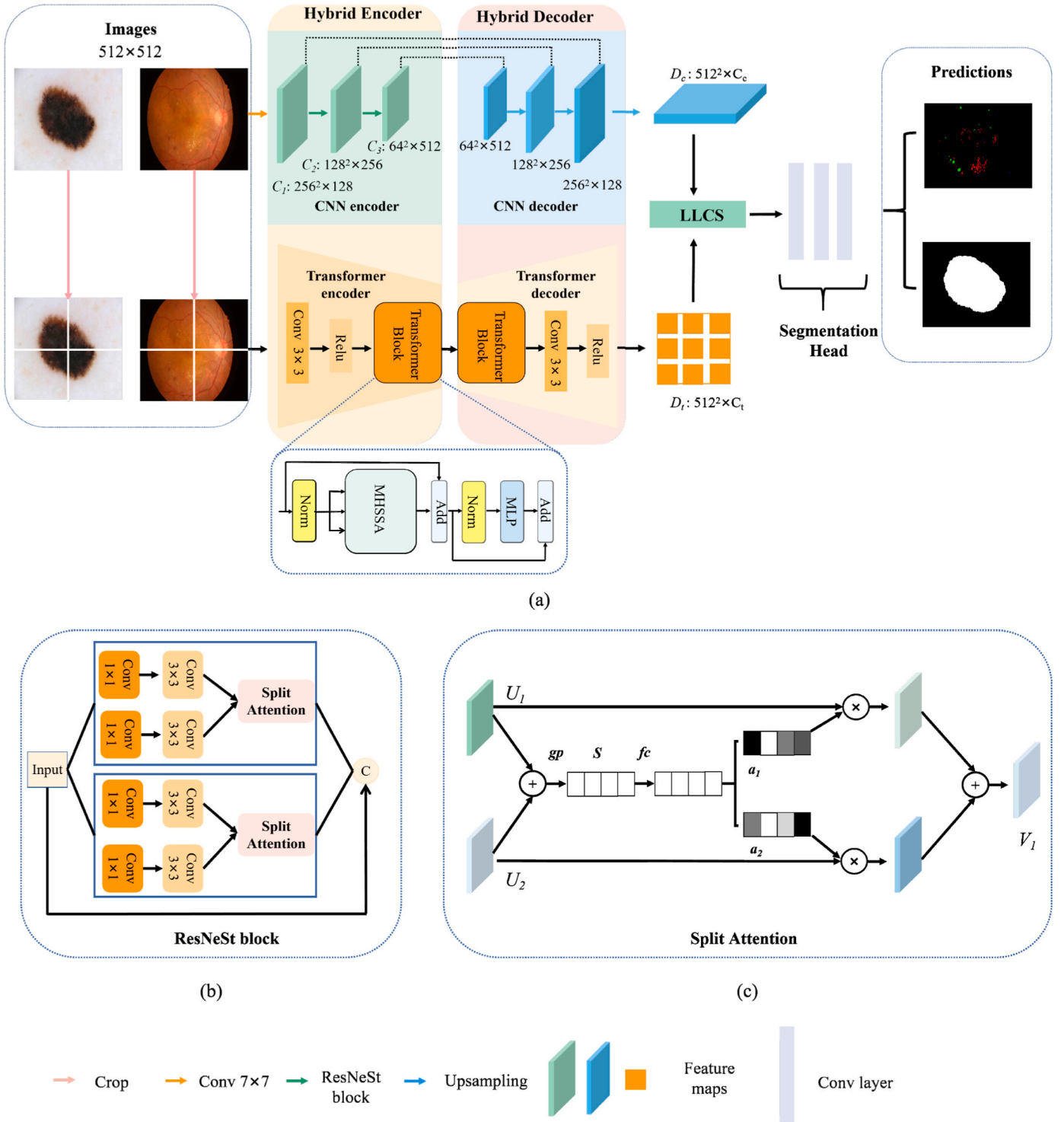


Fig. 2. (a) Overview of our proposed TC-Net. It consists of a hybrid encoder, a hybrid decoder, the locality-aware and long-range dependency concatenation strategy (LLCS) and a segmentation head. (b) The ResNeSt block. (c) The split attention module.

capabilities of capturing local features, which is essential for the network to recognize different lesions more accurately. In the segmentation of fundus lesions, Transformers tend to misclassify local lesions. In the example shown in Fig. 1 (b), the Vision Transformer misclassifies hard exudate (EX) pixels as soft exudate (SE) pixels. For local lesions, the segmentation result of CNN is better than that of Transformers, as shown in Fig. 1 (b). Therefore, it is also necessary to adaptively capture local information in medical image segmentation methods.

To overcome the above drawbacks, we present a hybrid architecture

named TC-Net to extract both local information and long-range dependencies simultaneously in medical images. First, TC-Net contains two branches: the CNN branch operates on downsampled medical images to retrieve the features of local regions, and the Transformer branch processes image patches to model the long-range context information among the regions. As shown in Fig. 1 (a), compared with the CNN-based method, our method can model both global and long-range context features and thus can differentiate the background pixels. In terms of modeling local information, our method can better capture the

fine-grained lesion information, which helps to distinguish multiple lesions. As shown in Fig. 1 (b), in contrast to the Transformer, our method can accurately segment local lesions. To concatenate the features derived from the two branches, inspired by DANet [16], we propose a locality-aware and long-range dependency concatenation strategy (LLCS). By fusing the outputs of the two branch modules, the LLCS block can focus on both local features and long-range dependencies in medical images to accomplish complementary fusion. Finally, the complementary feature maps obtained from the LLCS block are input into a segmentation head to produce confidence scores for each mask. Our main contributions can be summarized as follows:

- 1) We present a novel medical image method that simultaneously utilizes local information and long-range dependency. To achieve this idea, a hybrid structure including a CNN branch and a Transformer branch is proposed to extract regional features and model long-range context simultaneously.
- 2) We proposed a locality-aware and long-range dependency concatenation strategy, which design three parallel paths from the view of the channel and spatial attention to combining effectively the features obtained from the two branches.
- 3) We proposed a dynamic cyclical focal loss to relieve the class-imbalance problem in multi-lesion medical image segmentation.

2. Related work

Early semantic segmentation methods utilized feature engineering and machine learning techniques [17–19]. With the development of deep learning algorithms, semantic segmentation has achieved outstanding progress. Recently the state-of-the-art deep-learning-based image segmentation methods can be mainly categorized into two types: CNN-based methods and Vision Transformers.

CNN-based method. Long et al. [20] first designed an end-to-end convolutional network architecture for high-quality semantic segmentation. Inspired by Ref. [20], the more elegant U-Net [8] architecture was later proposed for biomedical image segmentation. U-Net uses a skip-connection strategy to integrate low-level and high-level features, thereby preserving some detailed features. In subsequent work, this U-shaped framework was taken as the basis for proposing various semantic segmentation methods. Res-UNet [21] was developed by introducing a weighted attention mechanism and the concept of residual connections into U-Net. Around the same time, a U-shaped network with nested and dense skip connections was proposed for multi-organ segmentation [22]. However, these methods collect features only from neighborhood pixels and lack awareness of the global information in images. In addition, ICNet [23] was proposed as a cascaded network architecture that concatenates multiresolution feature maps under proper label guidance and model compression. GLNet [24] was designed by combining two CNN branches to preserve both the global context and detailed local information; these branches interact with each other through a deep feature map sharing strategy, yielding better performance. However, such methods usually have limitations in modeling long-range relations because the convolution kernels tends to extract only local regions.

To solve this problem, some methods [25,26] have been proposed in which strided convolution is replaced with dilated convolution to enlarge the receptive field and model nonlocal information. Moreover, attention mechanisms have been proposed to help CNNs focus on long-range dependencies. For example, Wang et al. [27] applied self-attention modules in deep neural networks to explore the effectiveness of nonlocal dependencies in images, sequences, and videos. CBAM [28] and DANet [16] use attention mechanisms along the channel dimension and the spatial dimension, respectively, to strengthen the representation power of CNNs. CS-Net [29] integrates channel and spatial attention blocks into an FCN for the segmentation of curvilinear structures in medical images. These methods achieve remarkable

performance on different tasks. However, these methods are still influenced by the intrinsically local nature of convolution operations and consequently have limitations in modeling long-range dependencies.

Vision Transformers. Transformers [30] were initially proposed in the NLP domain and have achieved state-of-the-art performance in machine translation (ML). Inspired by their tremendous success in NLP tasks, many researcher have investigated the adaptation of Transformers for computer vision and medical image analysis tasks. SETR [31] provides a new perspective in which semantic segmentation is reconsidered as a sequence-to-sequence prediction problem with Transformers. Xie et al. [32] integrated the encoder-decoder architecture into Transformers and proposed an elaborate transparent object segmentation algorithm. The authors of ViSTR [33] applied the Transformer architecture for video segmentation tasks and achieved the highest speed among all methods considered for comparison. For medical image segmentation, TransUNet [15] utilizes a Transformer as an encoder to explore the global context information in medical images. Moreover, Hu Cao et al. [12] proposed a pure Transformer with an encoder-decoder structure for multi-organ segmentation. Transformers exhibit a powerful ability to model long-range dependencies. However, their performance is limited by a lack of ability to capture local features. Some recently relevant methods were applied to fuse the CNNs and Transformer into a new framework. Li et al. [34] present a dual encoding-decoding framework for medical image segmentation. However, they ignore exploring the difference between the Transformer and CNNs. Moreover, Xu et al. [35] proposed a feature extraction network, which composes the depthwise separable convolution and Transformer to segment the brain tumour and cardiac segmentation. Zhu et al. [36] proposed a medical segmentation approach that combines deep semantic features and edge information. Then, a feature fusion module was proposed to merge the semantic and edge features.

3. Methodology

As depicted in Fig. 2, the proposed TC-Net consists of four components: a hybrid encoder module, a hybrid decoder module, an LLCS block, and a segmentation head. The hybrid encoder consists of a CNN encoder and a Transformer encoder to capture the local information and long-range dependencies, respectively, in medical images. The hybrid decoder is adopted to recover spatial details from the high-level semantic features extracted by the hybrid encoder. Then, the LLCS is applied to concatenate the feature maps obtained from the hybrid decoder. Finally, the segmentation head is introduced as a classifier to obtain the final segmentation results.

3.1. The hybrid encoder

The hybrid encoder is built with a two-branch structure, including a CNN encoder and a Transformer encoder, to mine the local information and the long-range dependencies in medical images separately.

The CNN encoder. The CNN encoder includes three encoder layers, as shown in Fig. 2. In practice, we employ the pretrained ResNeSt block [37] as the backbone of this module. The ResNeSt block originated from ResNet [38] and integrates a split attention mechanism on different branches to capture cross-feature interactional information. Specifically, given an image $I \in R^{H \times W \times 3}$ with 3 color channels, where H and W are the height and width, the first encoder layer is applied to obtain the output feature map C_1 . The first encoder layer consists of a convolutional layer and a batch normalization (BN) layer followed by a nonlinear activation function (ReLU) and 3×3 max pooling with a stride of 2. The feature map C_1 is equally divided into K cardinal groups along the channel dimensions. Subsequently, each cardinal group is split into R individual groups, meaning that the feature map C_1 is divided into $G=KR$ groups in total. The K cardinal groups are fed into K parallel cardinal branches {cardinal 1, cardinal 2, ..., cardinal K }, and each parallel cardinal branch consist of R split branches {split 1, split 2, ..., split R } with a split

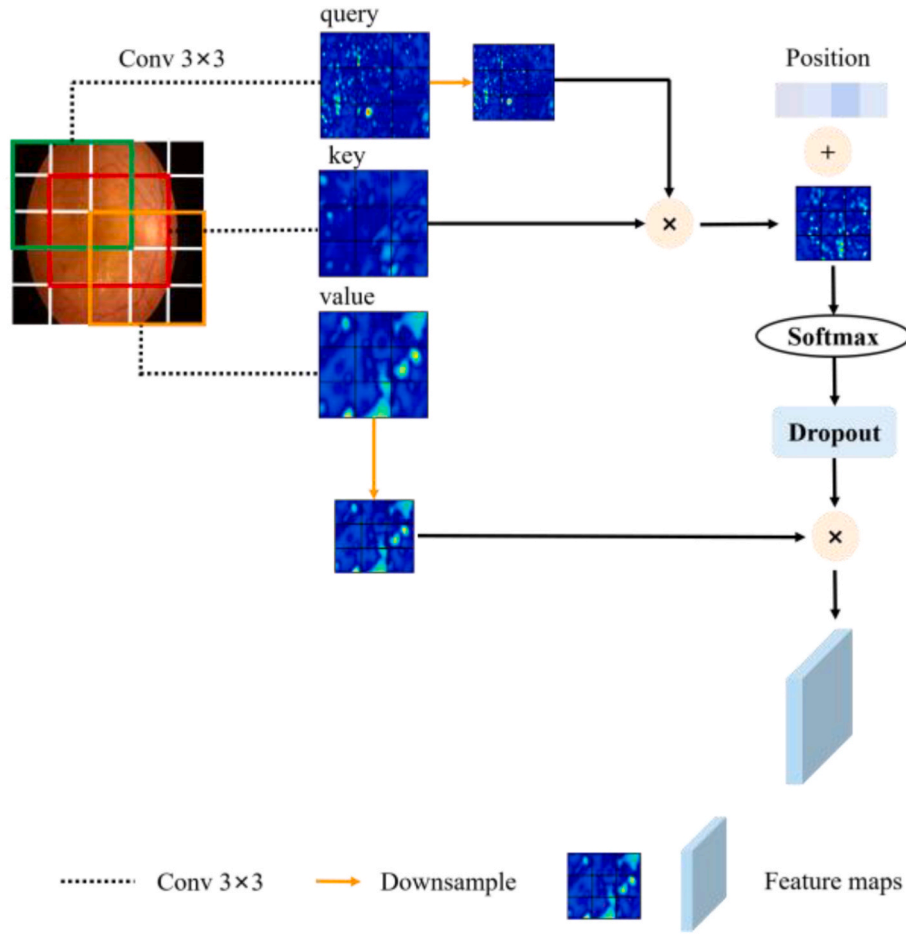


Fig. 3. The structure of the multihead squeezable self-attention (MHSSA) mechanism.

attention mechanism, as shown in Fig. 2 (b). Each split branch consists of two convolutional layers followed by BN and a ReLU activation function and outputs a feature map with height $H/4$ and width $W/4$ dimensions.

A split attention module is integrated into each cardinal branch to concatenate the R feature maps, denoted by $\{U_1, U_2, \dots, U_R\}$, that are derived from all split branches, as shown in Fig. 2 (c). This module first fuses the R feature maps via an element wise summation operation, and a global pooling operation is applied to obtain the global context information s :

$$s = \frac{1}{\frac{H}{4} \times \frac{W}{4}} \sum_{i=1}^{\frac{H}{4}} \sum_{j=1}^{\frac{W}{4}} [U_1(i, j) + U_2(i, j) + \dots + U_R(i, j)] \quad (1)$$

After the global pooling layer, two consecutive fully connected (FC) layers with a nonlinear activation function (softmax) are applied to predict the R attention weights $\{a_1, a_2, \dots, a_R\}$ on every split branch. Each attention weight is then used to perform the weighted fusion of $\{U_1, U_2, \dots, U_R\}$ to output one cardinal feature map V :

$$V = \sum_{j=1}^R a_j U_j \quad (2)$$

Correspondingly, we obtain K cardinal feature maps, denoted by $\{V_1, V_2, \dots, V_K\}$. The K cardinal feature maps are then concatenated along the channel dimension, and 1×1 convolutional layers with shortcut connections are used to produce the final ResNeSt block outputs C_2 :

$$C_2 = f^{1 \times 1}(\text{Concat}\{V_1, V_2, \dots, V_K\}) + T(C_1) \quad (3)$$

where $f^{1 \times 1}$ is a convolutional layer with kernel size 1×1 and T is an appropriate convolutional operation. Specifically, we set the hyper-parameters K and R to 2 and obtain two ResNeSt block outputs: $C_2 \in R^{\frac{H}{4} \times \frac{W}{4} \times 256}$ and $C_3 \in R^{\frac{H}{8} \times \frac{W}{8} \times 512}$.

The Transformer encoder. The Transformer encoder consists of one basic block and four Transformer blocks. The basic block receives image patches and obtains the feature map X . The basic block consists of a series of convolutional layers, BN, and 3×3 max pooling. Then, the feature map X is fed into the four Transformer blocks to capture the long-range dependencies. Specifically, the Transformer block includes a residual block and a Transformer module. The residual block transforms the feature map X into an intermediate feature map X_1 . Semantic segmentation requires dense prediction at the pixel level to classify each pixel as background or foreground. It is challenging to apply the Transformer architecture in image semantic segmentation due to its time complexity. In this paper, inspired by Ref. [39], we introduce multihead squeezable self-attention (MHSSA) mechanism instead of the multihead self-attention (MHSA) mechanism originally used in Transformers, as shown in Fig. 3.

We apply the MHSSA mechanism to obtain an attention matrix corresponding to the feature map. In this paper, we employ four attention heads, and the multihead mechanism is not presented in the formulation or the figure for simplicity. Concretely, three 3×3 convolution operations with different convolution kernels are applied to project X_1 to *query* (Q), *key* (K) and *value* (V) matrices: $Q = W_q X_1$, $K = W_k X_1$ and $V = W_v X_1$, the W_q , W_k and W_v denote the learnable parameters of the linear transformations. Then, the Q , K and V are flattened and transposed into sequences to compute the MHSSA outputs. In practice, to reduce the computational complexity, we apply a bilinear

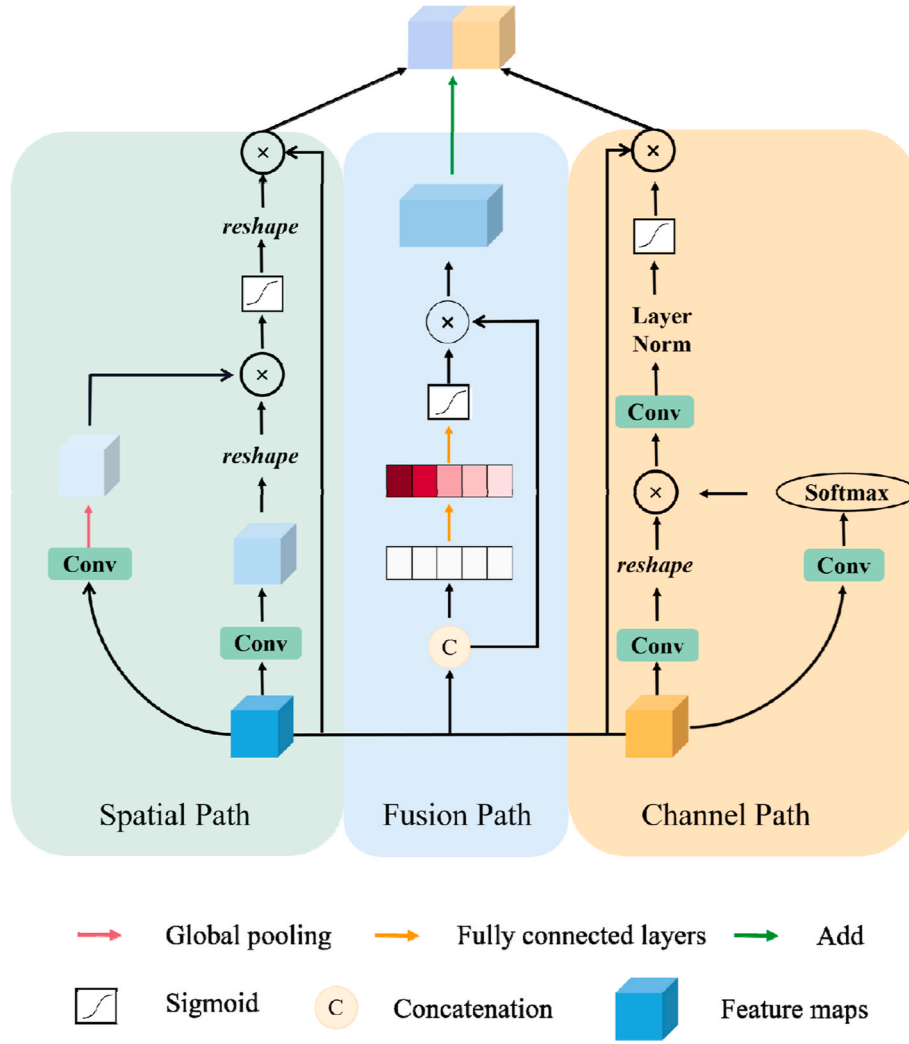


Fig. 4. Sketch of the LLCS block. It consists of three sub-paths: the spatial path, the channel path and the fusion path.

interpolation operation to downsample K and V , and add relative position information, followed by a Swin-Transformer [40].

$$MHSSA(Q, K, V) = \text{Softmax} \left(\frac{QBL(K^T)}{\sqrt{d}} + B \right) BL(V) \quad (4)$$

where BL is the bilinear operation, d is the dimensionality of Q/K , $\sqrt{d}$ is the appropriate normalization factor and B is the relative position bias.

Finally, we obtain the output of the Transformer block after passing through a multilayer perceptron (MLP) and a residual connection. And the final output obtained by the Transformer encoder is a series of feature maps $\{X_2, X_3, X_4, X_5\}$

3.2. The hybrid decoder

Corresponding to the hybrid encoder, the hybrid decoder also consists of two branches, namely, a CNN decoder and a Transformer decoder, as shown in Fig. 2 (a).

The CNN decoder. The CNN decoder consists of three blocks. Each block includes a bilinear upsampling function with a scale factor of 2 and a set of convolutional layers that halve the channel dimensions of the feature map. The feature map C_3 from the CNN encoder first undergoes an upsampling and convolution operation to transform it into an intermediate feature map with height $H/4$ and width $W/4$. Then, this intermediate feature map is concatenated with the low-level feature map C_2 . Finally, two 3×3 convolutional layers followed by BN and

ReLU are applied to the concatenated feature maps to reduce their channel dimensions.

The Transformer decoder. The Transformer decoder consists of four Transformer blocks and a skip connection to preserve fine-grained information. Unlike in the Transformer encoder, we apply the MHSSA mechanism to obtain an attention matrix to merge the feature representations from the low-level and high-level features. Firstly, we apply a 3×3 convolutional layers to the high-level feature map X_5 to compute the key (K) and value (V) matrices, while we obtain the query (Q) matrix from the low-level feature map X_4 : $K = W_k X_5$, $V = W_v X_5$ and $Q = W_q X_4$. Second, we compute the relation matrix between the low-level feature X_4 and the high-level feature map X_5 , followed by Eq.(4). Finally, we obtain the final output of the Transformer decoder.

3.3. Locality-aware and long-range dependency concatenation strategy (LLCS)

The Transformer needs to flatten the feature map to compute the attention matrix, which causes channel information to be lost to some degree. Meanwhile, the CNN model processes one neighborhood region at a time, which is insufficient to capture the nonlocal features. Therefore, given the features $D_c \in R^{H \times W \times C_c}$ from the CNN decoder and the features $D_t \in R^{H \times W \times C_t}$ from the Transformer decoder, we fuse the local and long-range dependency information along the channel and spatial dimensions. As shown in Fig. 4, the proposed LLCS block consists of

Table 1

Statistical information of the IDRiD and DDR datasets, where the values represent the percentages of multiclass pixels among all pixels.

Dataset	BK	EX	HE	MA	SE
IDRiD	97.8%	0.90%	1.01%	0.10%	0.19%
DDR	99.41%	0.19%	0.33%	0.02%	0.05%

three paths: the spatial path, the channel path and the fusion path.

Spatial path. As depicted in Fig. 4, we first feed the $D_c \in R^{H \times W \times C_c}$ into two 1×1 convolutional layers and generate two feature maps: $D_c^1 \in R^{H \times W \times \frac{C_c}{2}}$ and $D_c^2 \in R^{H \times W \times \frac{C_c}{2}}$. Then, global pooling is applied to D_c^1 :

$$D_c^3 = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W D_c^1(\cdot, i, j) \quad (5)$$

In this way, two feature maps D_c^2 and D_c^3 are reshaped to $R^{N \times \frac{C_c}{2}}$, where $N = H \times W$. We fuse the transposes of D_c^3 and D_c^2 by means of a matrix dot product operation followed by a sigmoid layer to output the attention map S_a :

$$S_a(i, j) = \frac{\exp(D_c^3(i, \cdot) D_c^2(\cdot, j))}{\sum_{i=1}^N \exp(D_c^3(i, \cdot) D_c^2(\cdot, j))} \quad (6)$$

Finally, we multiply S_a by the input D_c to obtain the output of the spatial path, O_s .

Channel path. The channel path receives the feature map D_t as input and applies two 1×1 convolution operations to it. Thus, we obtain the two feature maps: $D_t^1 \in R^{H \times W \times \frac{C_t}{2}}$ and $D_t^2 \in R^{H \times W \times 1}$. Then, we reshape D_t^1 and D_t^2 to $R^{N \times C}$, $R^{N \times 1}$, receptively. We apply the softmax function to D_t^2 to obtain the channel attention matrix S_c . Finally, we obtain the channel weights W_c based on the channel attention matrix:

$$W_c = \sigma(LN(f^{1 \times 1}(D_t^1 \cdot S_c))) \quad (7)$$

where σ denotes the sigmoid activate function, $f^{1 \times 1}$ is a convolutional layer with a filter size of 1×1 and LN represents the layer norm operation. We multiply W_c by the input D_t to obtain the final output of the channel path, O_c .

Fusion path. We design a fusion path to enhance the feature representation quality by modeling the channel relationship between the features from the CNN decoder and the Transformer decoder. We fuse the feature maps D_c and D_t by means of a channelwise concatenation operation and use global average pooling to generate the global spatial representation in the form of a matrix z_c . Then, two fully connected layers followed by a sigmoid activation are applied to z_c . Finally, the output of the fusion path is $O_f = z_c \cdot \text{Concat}\{D_c, D_t\}$, where $\cdot$ denotes

channelwise multiplication.

Ultimately, we fuse the three feature maps in accordance with the following formulation to obtain the final output O .

$$O = \text{Concat}\{O_s, O_c\} + O_f \quad (8)$$

3.4. The segmentation head

At the end of TC-Net, a segmentation head is designed to compute the confidence scores for each class. The input to the segmentation head is the feature map O from the LLCs block. The segmentation head consists of two layers: the first layer is a 3×3 convolutional layer followed by the ReLU function, and the second layer is a 1×1 convolutional layer to obtain the final output of TC-Net, $y_{pred} \in R^{H \times W \times c}$ (where c is the total number of classes).

3.5. The loss function

Lin et al. [41] proposed the focal loss to strike a suitable balance between the foreground and background classes in one-stage object detection during training. When the number of training samples is imbalanced, the focal loss demonstrates superior performance. Based on the focal loss, Smith et al. [42] proposed the cyclical focal loss (CFL), which has a better generalization ability than the original focal loss. However, in our task, the distribution of the foreground and background pixels is extremely unbalanced. Moreover, imbalance is also evident among the different foreground categories, as shown in Table 1. Therefore, the CFL is also not suitable for our segmentation tasks. Inspired by the CFL, we proposed the dynamic cyclical focal loss (DCFL) to train the segmentation model:

$$DCFL = I(e_i, e_n) F_{hc} + \beta(1 - I(e_i, e_n)) F_{lc} \quad (9)$$

where β is a dynamic weight calculated for each class in every epoch given the ground truth.

$$\beta = \left[\log \left(1.10 + \frac{\sum_{i=1}^{H \times W} Y^{(i)} = k}{\sum_{k=1}^c \sum_{i=1}^{H \times W} Y^{(i)} = k} \right) \right]^{-1}, k \in [1, c] \quad (10)$$

where c is the total number of total classes, H and W denote the dimensions, and $Y^{(i)}$ represents the ground truth for pixel i . $I(e_i, e_n)$ denotes a function that varies with the training epoch:

$$I(e_i, e_n) = \begin{cases} 1 - r \frac{e_i}{e_n} & \text{if } r \times e_i \leq e_n \\ \left(r \frac{e_i}{e_n} - 1 \right) / (r - 1) & \text{otherwise} \end{cases} \quad (11)$$

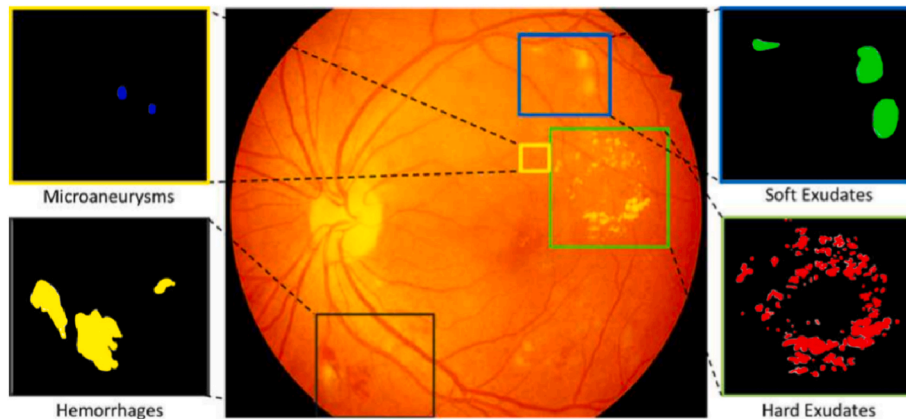


Fig. 5. Examples of the typical lesions in fundus images from the IDRiD dataset.

Table 2

Comparisons of multi-lesion segmentation performance on the IDRiD dataset (best results are shown in bold).

Methods	EX		MA		SE		HE		ALL			
	AUC	pixel acc	AUC	pixel acc	AUC	pixel acc	AUC	pixel acc	mAcc	mIoU	mKappa	mDice
U-Net	0.8927	0.6160	0.8053	0.4705	0.6548	0.6993	0.7822	0.5356	0.6627	0.5437	0.6002	0.6766
DeepLabv3+	0.9851	0.7630	0.9376	0.2329	0.9377	0.4250	0.9183	0.4037	0.5640	0.4993	0.5525	0.6229
L-Seg	-	0.7370	-	0.2942	-	0.3497	-	0.4657	0.5684	0.5102	0.5692	0.6393
X-Net	0.9829	0.6318	0.8624	0.2596	0.9699	0.3996	0.9437	0.4766	0.5525	0.4846	0.5359	0.6129
Y-Net	0.9280	0.4750	0.7593	0.1747	0.9108	0.5072	0.7701	0.3343	0.4970	0.4278	0.4481	0.5490
Swin-UNet	0.9735	0.4671	0.8633	0.0279	0.9873	0.5413	0.9507	0.4326	0.4921	0.4008	0.4095	0.5065
Zhu et al.	0.9447	0.7710	0.8068	0.2879	0.8198	0.3246	0.8826	0.4610	0.5676	0.4855	0.5387	0.6100
Ours	0.9948	0.7682	0.9881	0.3669	0.9985	0.7731	0.9818	0.5905	0.6985	0.5696	0.6387	0.6968

where r is a cyclical factor, e_t represents the number of the current training epoch, and e_n represents the total number of training epochs. The function F_{hc} is designed to concentrate on samples with higher confidence at the beginning of training, and F_{lc} is the focal loss:

$$F_{hc} = -(1 + p_t)^{\gamma_1} \log(p_t) \quad (12)$$

$$F_{lc} = -(1 - p_t)^{\gamma_2} \log(p_t) \quad (13)$$

where p_t is the softmax probability of each class, γ_1 and γ_2 are focusing parameters.

4. Experimental results

4.1. Datasets for the experiments

We tested the proposed TC-Net on two different applications: multi-lesion segmentation of fundus images and skin lesion segmentation. Two databases, IDRiD [43] and DDR [44], were employed in the multi-lesion segmentation experiments. IDRiD (Indian Diabetic Retinopathy Image Dataset) (Porwal et al. (2018)) was acquired by a retinal specialist at an eye clinic located in Nanded, Maharashtra, India. The images were captured using a Kowa VX-10 alpha digital fundus camera with a 50-degree field of view (FOV), and all are centered near the macula. The IDRiD dataset contains a total of 81 images with dimension 4288×2848 , each of which are provided with pixel-level annotations for microaneurysms (MA), hard exudate (EX), soft exudate (SE) and hemorrhage (HE), as shown in Fig. 5. The DDR dataset consists of 13673 retinal images obtained at 147 hospitals from 23 provinces in China. This dataset was collected using 42 types of fundus cameras with a 45° FOV. For the multi-lesion segmentation task, 757 fundus images were selected, in which lesions including EX, MA, SE, and HE are annotated. In our experiments, we used 483 images to train the model and the remaining 274 images for testing.

The second application is skin lesion segmentation. We used the ISIC-2017 dataset [45] to evaluate our proposed method. This dataset contains 2750 dermoscopic images, and a ground-truth is provided for each image. For our task, we selected 2000 images for training and reserved the remaining 750 images for testing.

4.2. Implementation details

The proposed method was implemented on the public deep learning platform PyTorch [46]. All the experiments were performed on an Ubuntu 18.04 system with an NVIDIA RTX 3090 GPU with 24 gigabytes of memory. During training, all images were resized to 512×512 , and a batch size of 4 was used. The model was trained using the Adam optimizer algorithm with an initial learning rate of 5×10^{-4} and a weight decay of 0.00001. Empirically, we set r to 2, γ_1 to 3 and γ_2 to 2 for the DCFL. Moreover, data augmentation was applied in our task, including random horizontal flipping, color transformation and random affine transformation.

4.3. Multi-lesion segmentation of fundus images

We first evaluated the proposed TC-Net on the multi-lesion segmentation of fundus images. Two databases, IDRiD and DDR, were used in the experiments. We compared TC-Net with state-of-the-art segmentation algorithms, including U-Net, DeepLabv3+ [47], L-Seg [48], X-Net [34], Y-Net [49], Swin-UNet and Zhu et al. [36].

To evaluate the multi-lesion segmentation results, the following metrics are used and compared in our paper:

- Area Under the ROC Curve (AUC);
- Pixel accuracy (pixel acc) = $\frac{\sum_i n_{ii}}{\sum_i t_i}$;
- Intersection over union (IoU) = $\frac{TP}{TP+FP+FN}$;
- Kappa score = $\frac{(Accuracy - p_e)}{1 - p_e}$;
- Dice = $\frac{2 \times TP}{(FP+FN+2 \times TP)}$;

where n_{ij} is the number of pixels of class i predicted to belong to class j , $t_i = \sum_j n_{ij}$, $Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}$, TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. And p_e illustrates opportunity consistency between the ground truth and prediction:

Table 3

Comparisons of multi-lesion segmentation performance on the DDR dataset (best results are shown in bold).

Methods	EX		MA		SE		HE		ALL			
	AUC	pixel acc	AUC	pixel acc	AUC	pixel acc	AUC	pixel acc	mAcc	mIoU	mKappa	mDice
U-Net	0.6426	0.4023	0.6122	0.3580	0.8014	0.4453	0.6926	0.2683	0.4944	0.3849	0.3872	0.4981
DeepLabv3+	0.9325	0.3675	0.9003	0.1427	0.8827	0.2586	0.7473	0.2340	0.4004	0.3693	0.3572	0.4757
L-Seg	-	-	-	-	-	-	-	-	-	-	-	-
X-Net	0.9753	0.4087	0.8491	0.1801	0.9651	0.5101	0.9627	0.3372	0.4869	0.3945	0.4012	0.5079
Y-Net	0.7677	0.3126	0.7053	0.2192	0.8045	0.4710	0.6711	0.1766	0.4356	0.3369	0.3024	0.4329
Swin-UNet	0.9854	0.4135	0.8725	0.0590	0.9869	0.4071	0.9652	0.3237	0.4410	0.3814	0.3730	0.4849
Zhu et al.	0.9205	0.3759	-	-	0.9445	0.4379	0.8824	0.2586	0.4146	0.3537	0.3289	0.4447
Ours	0.9942	0.5129	0.9758	0.2261	0.9928	0.4675	0.9810	0.3817	0.5171	0.3969	0.4070	0.5092

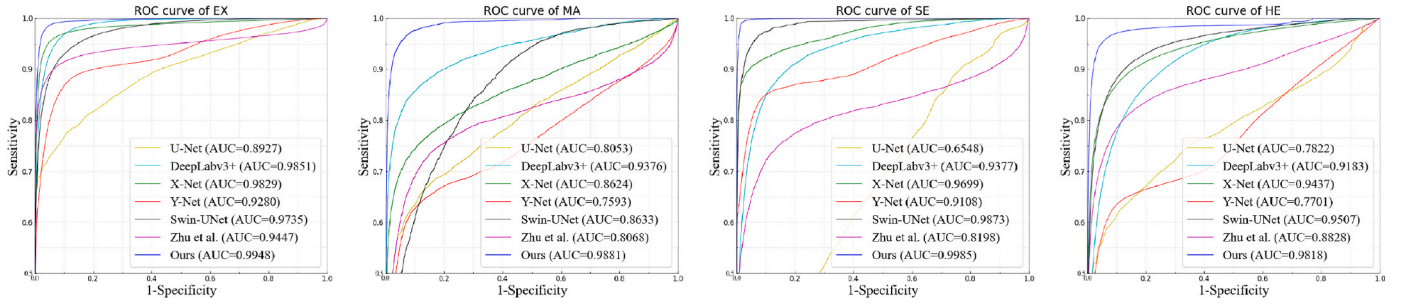


Fig. 6. The ROC curves with AUC scores for EX, MA, SE and HE on the IDRiD dataset.

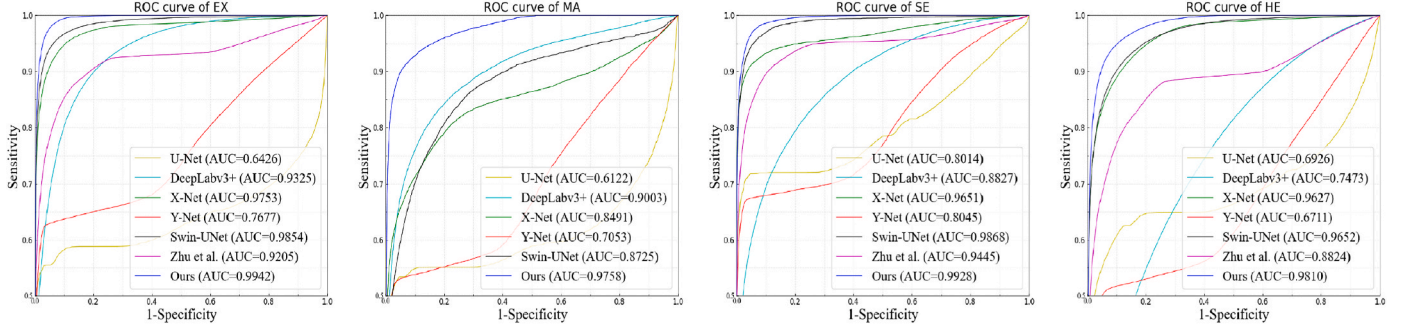


Fig. 7. The ROC curves with AUC scores for EX, MA, SE and HE on the DDR dataset.

$$p_e = \frac{((TP + FN)(TP + FP) + (TN + FP)(TN + FN))}{(TP + FN + FP + FN)^2} \quad (14)$$

Accuracy analysis results. The evaluation scores for each dataset are summarized in Table 2 and Table 3. Table 2 shows the comparison among these methods on the IDRiD database. As can be seen, TC-Net achieved AUC and pixel acc values of 0.9948 and 0.7682, respectively, for EX segmentation on the IDRiD database, better than those of

the other methods in the metric of AUC. Compared with DeepLabv3+, the AUC increased by 0.97%. And compared with Zhu et al. the AUC increased by 5.01%. For the SE segmentation results, compared with U-Net, the pixel acc increased from 0.6993 to 0.7731, a difference of 7.38%. In addition, the AUC and pixel acc values of the proposed method reached 0.9818 and 0.5905, respectively, for HE segmentation. Compared with U-Net, we achieved pixel acc improvements of 5.49%. Nevertheless, U-Net outperformed the proposed method in terms of the

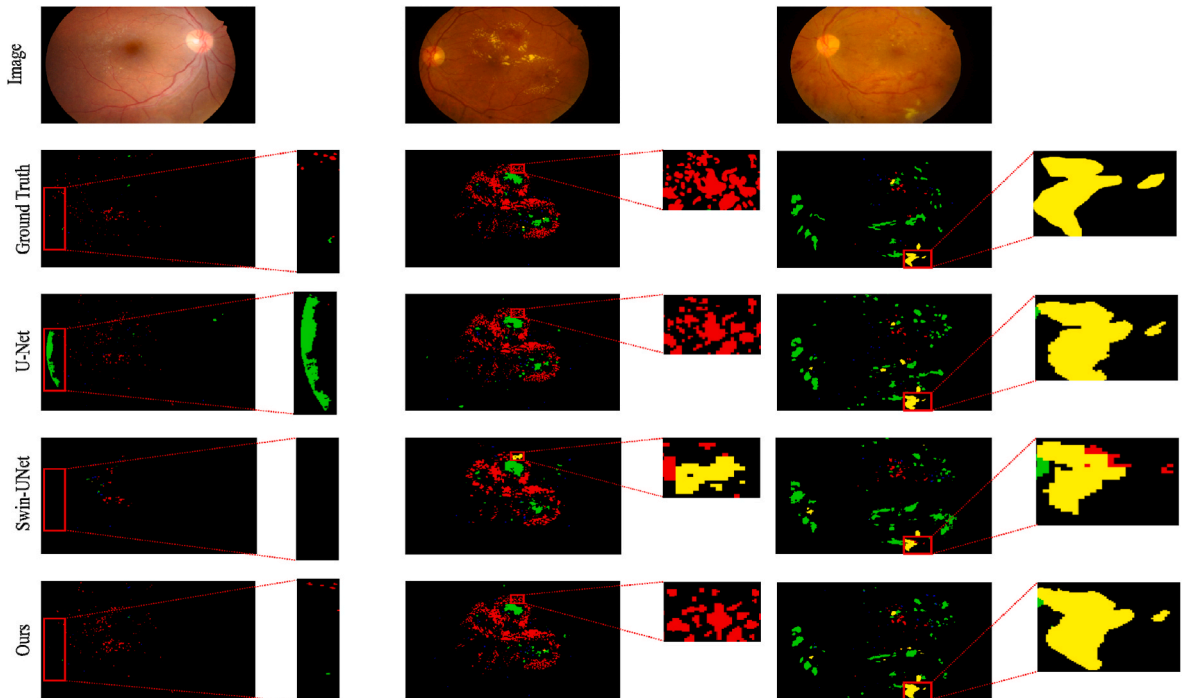


Fig. 8. Visual examples of multi-lesion segmentation in the IDRiD database, where the red markings denote EX, the blue markings denote MA, and the yellow and green markings denote SE and HE, respectively.

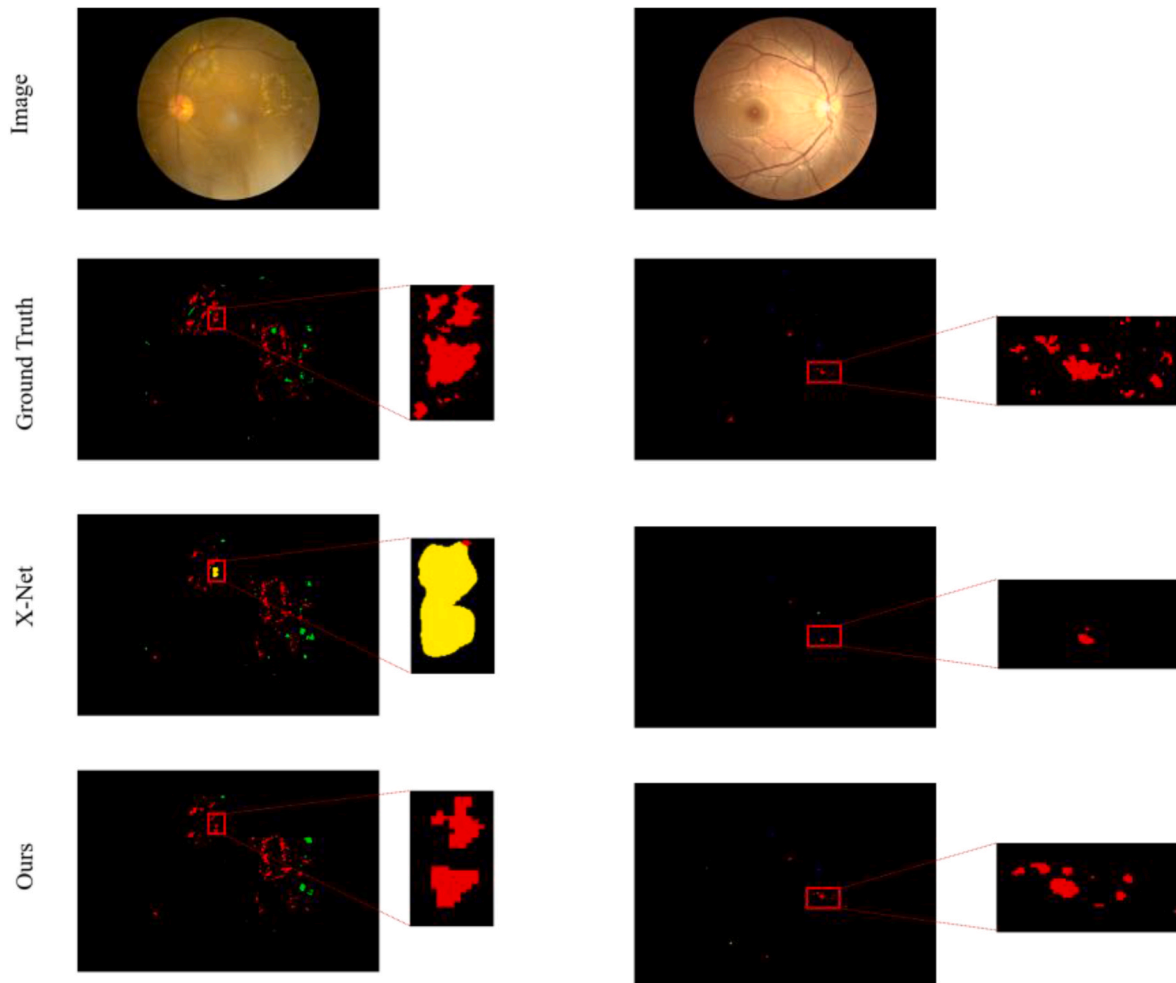


Fig. 9. Visual examples of multi-lesion segmentation in the DDR database, where the red markings denote EX, the blue markings denote MA, and the yellow and green markings denote SE and HE, respectively.

MA segmentation results. Here, we used the mean pixel accuracy (mAcc), mean IoU (mIoU), mean Kappa (mKappa) and mean Dice (mDice) values to comprehensively compare the segmentation results of U-Net and TC-Net. U-Net achieved mAcc, mIoU, mKappa and mDice scores of 0.6627, 0.5437, 0.6002 and 0.6766, respectively, lower than the corresponding scores of 0.6985, 0.5696, 0.6387 and 0.6968 achieved by TC-Net. Moreover, the proposed method achieved considerable improvements in the segmentation of EX, SE and HE compared with U-Net, as shown in Table 2.

The performance comparison on the DDR dataset is presented in Table 3. Our proposed method achieved AUC and pixel acc scores of 0.9942 and 0.5129 for EX segmentation on the DDR database, better than those of the other methods. It similarly achieved scores of 0.9758 and 0.2261 for MA segmentation. For HE segmentation, the proposed method achieved AUC and pixel acc scores of 0.9810 and 0.3817, respectively. Compared to Swin-UNet, our method showed considerable improvements of 1.58% and 5.44% in the AUC and pixel acc metrics, respectively. Moreover, compared with the second-best method, X-Net, the mAcc increased from 0.4869 to 0.5171, a difference of 3.02%, the mIoU increased from 0.3945 to 0.3969, a difference of 0.24%, the mKappa improved by 0.58% and the mDice improved by 0.13%.

To further compare the multi-lesion segmentation performance of the different approaches, the receiver operating characteristic (ROC) curves of these approaches on the two databases are shown in Fig. 6 and Fig. 7. It can also be observed from Fig. 6 and Fig. 7 that the proposed method achieved the highest AUC scores among all methods on IDRiD

and DDR datasets.

Visual analysis results. In fundus images, the size of most lesions is tiny, necessitating a segmentation network that can distinguish detailed information. Furthermore, the distribution of multiple lesions is typically scattered in fundus images, which makes it difficult to model long-range dependencies among lesions. In this scenario, the CNN-based method obtains higher scores than the Visual Transformer, as shown in Tables 2 and 3. However, the function of Transformers in multi-lesion segmentation cannot be ignored. Thus, we also present the multi-lesion segmentation results for fundus images from the IDRiD and DDR datasets to illustrate the importance of the combination of local information and long-range dependencies, as shown in Fig. 8 and Fig. 9.

As shown in Fig. 8, there are some errors in the segmentation between background and HE in the segmentation results derived by U-Net, reflecting its lack of ability to model long-range dependencies. In contrast, Swin-UNet can recognize the background in fundus images. However, we can observe from the multisegmentation results that Swin-UNet shows poor performance in small regions and exhibits serious misclassification for EX segmentation. This demonstrates that Swin-UNet poorly handles the local information in fundus images. Meanwhile, the proposed method can recognize not only background but also small lesions. The segmentation results show that our method achieves accurate segmentation.

On the DDR dataset, we compared the proposed method with the second-best method achieved by X-Net, as shown in Fig. 9. The segmentation results show that the X-Net exists misclassification between

Table 4

Skin lesion segmentation performance on the ISIC-2017 dataset compared with other methods (best results are shown in bold).

Method	mAcc	mIoU	mKappa	mDice
U-Net	0.7946	0.6658	0.5864	0.7932
DeepLabv3+	0.8898	0.7940	0.7664	0.8831
X-Net	0.8438	0.7692	0.7330	0.8659
Y-Net	0.8268	0.7225	0.6688	0.8343
Swin-UNet	0.8825	0.8270	0.8075	0.9034
Zhu et al.	0.8711	0.7956	0.7677	0.8837
Ours	0.8886	0.8290	0.8099	0.9048

EX and SE. Moreover, for the X-Net method, the segmentation is coarse in EX segmentation results. While the proposed method can capture a more accurate EX region than X-Net. And it is also easy to observe that the segmentation results of our proposed method are closer to the ground truth than X-Net. The segmentation results again show that our method generates better segmentation maps than the other methods. Thus, the effectiveness of combining CNN and Transformer models is demonstrated.

4.4. Skin lesion segmentation

We used the ISIC-2017 dataset to evaluate the performance of our method in the skin lesion segmentation task and compared the proposed

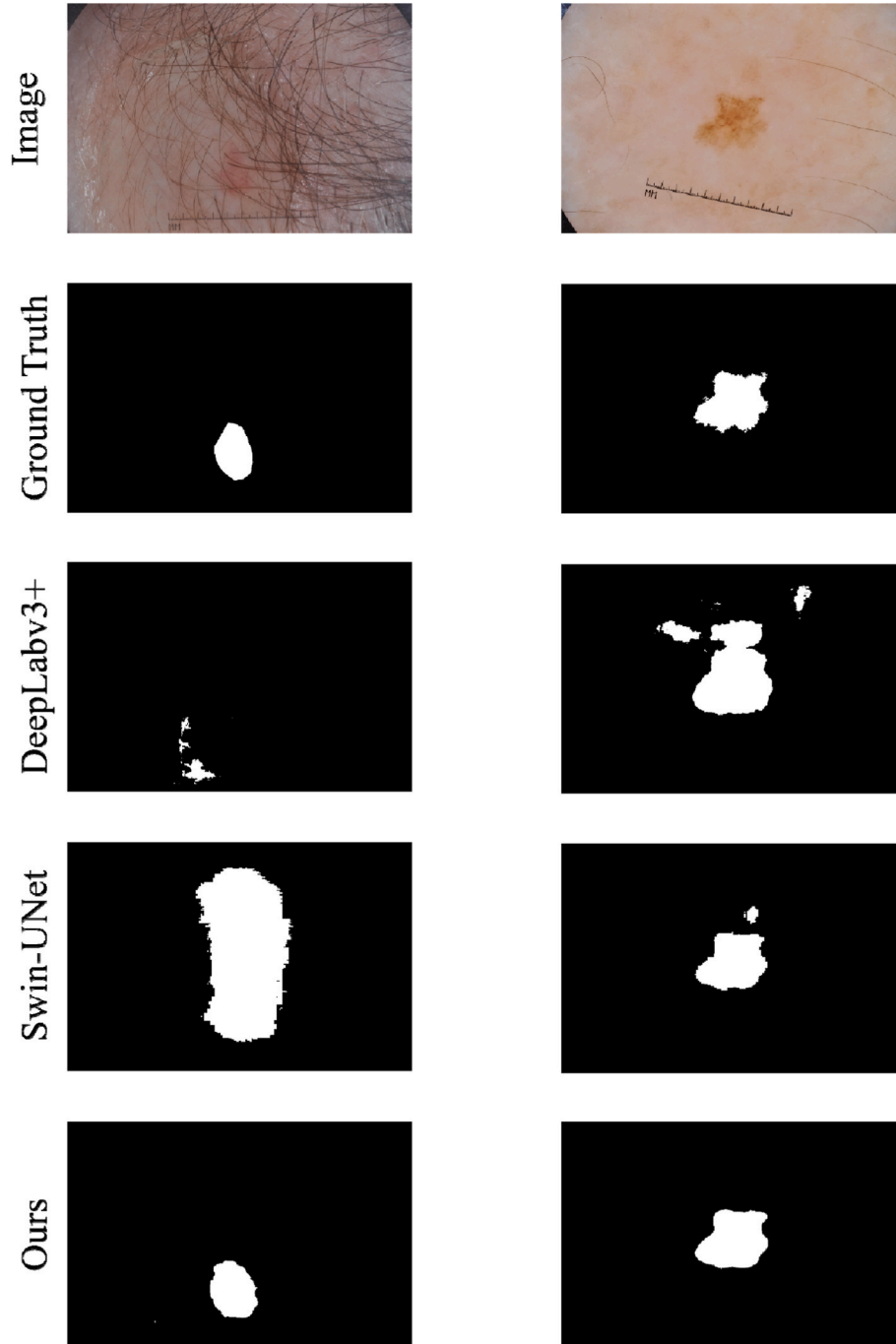


Fig. 10. Visual examples of skin lesion segmentation on the ISIC-2017 dataset.

Table 5

Ablation study results of the proposed method on the IDRiD dataset (best results are shown in bold).

Method	mAcc	mIoU	mKappa	mDice
CNN branch	0.6254	0.5511	0.6115	0.6788
Transformer branch	0.5310	0.4734	0.5205	0.6036
CNN + Transformer	0.6885	0.5300	0.5889	0.6600
TC-Net	0.6985	0.5696	0.6387	0.6968

method with several competitive methods, including U-Net, X-Net, Y-Net, Swin-UNet and Zhu et al. to evaluate the skin lesion segmentation performance.

Accuracy analysis results. The evaluation scores for skin lesion segmentation are shown in Table 4. This table shows that the proposed method achieved mAcc, mIoU, mKappa and mDice scores of 0.8886, 0.8290, 0.8099 and 0.9048, respectively, on the ISIC-2017 database. Compared with Swin-UNet, the mAcc increased from 0.8825 to 0.8886, a difference of 0.61%, the mIoU increased from 0.8270 to 0.8290, a difference of 0.2%, the mKappa improved by 0.24% and the mDice

improved by 0.14%, respectively, as shown in Table 4.

Visual analysis results. Visualizations of the segmentation results of the different approaches are shown in Fig. 10. As observed from Fig. 10, the DeepLabv3+ method results in serious misclassification for skin lesion segmentation. There are many errors in segmentation between background and skin lesion pixels. In particular, when other interfering factors (thick hairs) are present in the input images, DeepLabv3+ tends to produce coarse segmentation results, as shown in Fig. 10. In contrast, Swin-UNet is more suitable for skin lesion segmentation than DeepLabv3+. Nevertheless, the proposed method achieves more accurate skin lesion segmentation results than Swin-UNet. This finding suggests that our method is well adapted to the segmentation of medical images.

4.5. Ablation study

Ablation study on network's each module. We performed an ablation study on the IDRiD database, as shown in Table 5. In this section, we also use mAcc, mIoU, mKappa and mDice to comprehensively

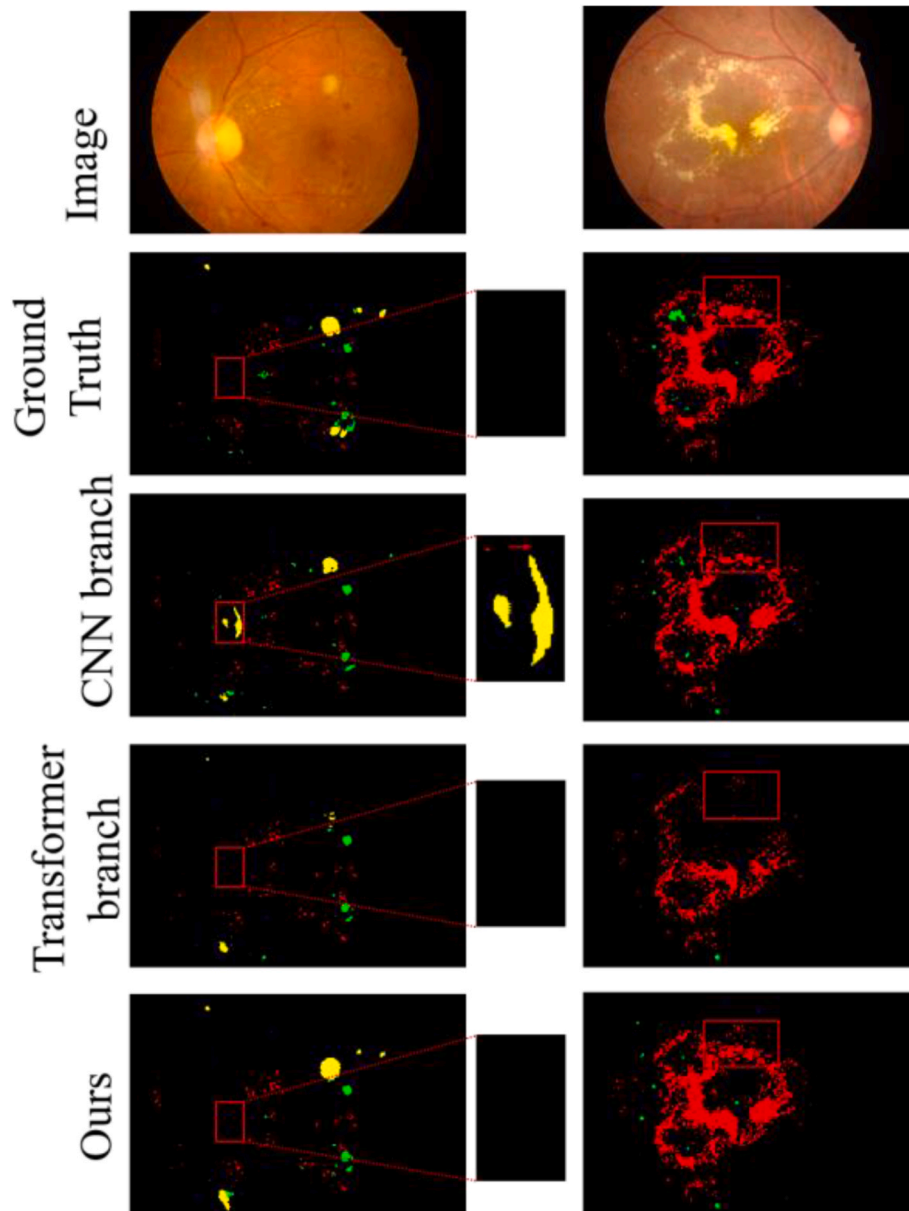
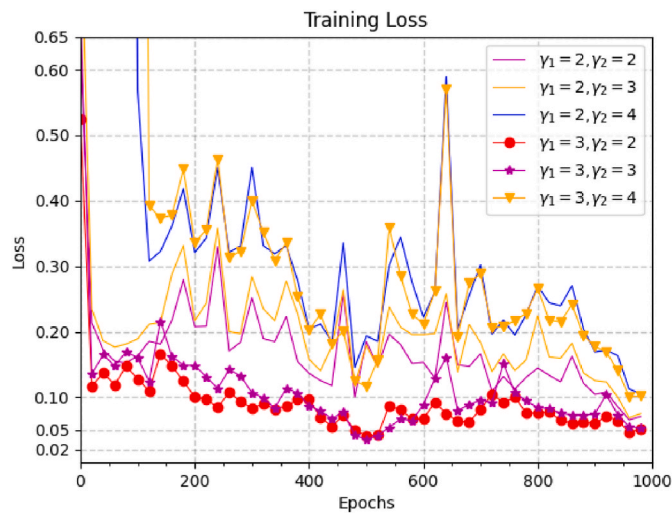


Fig. 11. Ablation study for multi-lesion segmentation on the IDRiD dataset.

Table 6

Ablation study results of the MHSSA on the IDRiD database (best results are shown in bold).

Methods	mAcc	mIoU	mKappa	mDice	GFLOPs	Params	Inference Time	GPU Memory
MHSA	0.7037	0.5737	0.6413	0.7012	430	51.7 M	150.0 ms	15259 MB
MHSSA	0.6985	0.5696	0.6387	0.6968	414	46.9 M	133.3 ms	13011 MB

**Fig. 12.** Comparison of the training loss across all training epochs when using different γ_1 and γ_2 values.

evaluate the performance of each module. The experimental results show that the CNN branch achieved mAcc, mIoU, mKappa and mDice scores of 0.6254, 0.5511, 0.6115 and 0.6788, respectively. In addition, the Transformer-branch achieved mAcc, mIoU, mKappa and mDice values of 0.5310, 0.4734, 0.5205 and 0.6036, respectively. When we combined the CNN and Transformer models, the combined model achieved mAcc, mIoU, mKappa and mDice values of 0.6885, 0.5300, 0.5889 and 0.6600, respectively. When the extracted feature maps were merged via the LLCs block, the mAcc improved by 1.0%, the mIoU improved by 3.96%, the mKappa increased by 4.98% and the mDice increased by 3.68%. We also present the multi-lesion segmentation results to illustrate the importance of each module, as shown in Fig. 11. We can be seen from Fig. 11 that the CNN-branch recognized lesions well, but there exists misclassification between the lesion and background. For the Transformer-branch, the segmentation is coarse due to its lack of local information. However, compared with the CNN branch and the Transformer branch, the TC-Net achieved better segmentation results. This ablation study proves the effectiveness of the improvements proposed in this study.

Ablation study on MHSSA. The performance comparison between MHSSA and MHSA is shown in Table 6. When we use the MHSA, it achieved 0.7037, 0.5737, 0.6413 and 0.7012 in terms of mAcc, mIoU, mKappa and mDice, respectively. However, the MHSSA used fewer parameters and FLOPs than the MHSA. As shown in Table 6, the MHSSA achieved 0.6985 mAcc with 46.9 M parameters and 414 G FLOPs which is smaller than the MHSA. Meanwhile, the MHSSA runs faster than the MHSA under the same hardware conditions. Our method requires about 13 GB GPU memory, when the batch size is 2. Nevertheless, training the network with MHSA requires over 15 GB of GPU memory. The above analysis shows that our method achieves accuracy-efficiency trade-offs compared to MHSA.

Ablation study on loss function. We compared the results obtained with different γ_1 and γ_2 values. Fig. 12 shows how the losses changed with various values of γ_1 and γ_2 throughout training epochs. The loss decreased faster with $\gamma_1 = 3$ and $\gamma_2 = 2$ than with the other settings. As displayed in Fig. 12, the loss converged to a value of 0.049 after 1000

Table 7

Performance comparison of different loss functions (best results are shown in bold).

Loss Function	mAcc	mIoU	mKappa	mDice
Dice Loss	0.6658	0.5640	0.6331	0.6930
CE Loss	0.5661	0.5230	0.5709	0.6434
Focal Loss	0.6197	0.5576	0.6208	0.6820
CFL	0.6427	0.5544	0.6171	0.6807
Ours	0.6985	0.5696	0.6387	0.6968

training epochs.

To evaluate the effectiveness of the DCFL function, we performed experiments with different loss functions on the IDRiD database. We first trained TC-Net using the Dice loss with the same learning rate strategy. Then, we replaced the Dice loss with the cross-entropy (CE) loss, the focal loss, and the CFL to train the network. As shown in Table 7, the proposed DCFL achieved better segmentation results than other compared loss functions in the presence of class imbalance.

5. Conclusion

In this work, we have proposed TC-Net, a novel end-to-end segmentation network that integrates the CNN and Transformer architectures into a unified architecture for medical image segmentation. To model local information and long-range dependencies simultaneously, we designed both a hybrid encoder and a hybrid decoder. Moreover, we proposed a novel feature fusion strategy, LLCs, for merging the different feature maps from the CNN and Transformer branches to obtain more delicate segmentation results. In addition, to solve the class-imbalance problem, we proposed the DCFL function. Finally, we evaluated TC-Net on two applications, namely, multi-lesion segmentation for fundus images and skin images. The experimental results prove that TC-Net performs better than other competitive segmentation methods. The proposed method is a general segmentation framework and can also be applied for the segmentation of other types of biomedical images.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yu Li reports financial support was provided by National Natural Science Foundation of China.

Acknowledgements

This work was supported by grants from National Key R&D Program of China 12 (2017YFC0909600); Scientific Research Project of Beijing Educational Committee (KM202110005024) and the Natural Science Foundation of China (41706201).

References

- [1] S. Fortunato, Community detection in graphs, *Phys. Rep.* 486 (3–5) (2009).
- [2] Arti Tiwari, Shilpa Srivastava, Millie Pant, Brain tumor segmentation and classification from magnetic resonance images: review of selected methods from 2014 to 2019, *Pattern Recogn. Lett.* 131 (2020) 244–260.
- [3] Mohd Ali Balafar, Abdul Rahman Ramli, M. Iqbal Saripan, Mashohor Syam- siah, Review of brain mri image segmentation methods, *Artif. Intell. Rev.* 33 (3) (2010) 261–274.

- [4] Johannes Hofmanninger, Florian Prayer, Jeanny Pan, Sebastian Rohrich, Helmut Prosch, Georg Langs, Automatic lung segmentation in routine imaging is primarily a data diversity problem, not a methodology problem, *European Radiology Experimental* 4 (1) (2020) 1–13.
- [5] Tomas Vicar, Balvan Jan, Josef Jaros, Florian Jug, Radim Kolar, Michal Masarik, Jaromir Gumulec, Cell segmentation methods for label-free contrast microscopy: review and comprehensive comparison, *BMC Bioinf.* 20 (1) (2019) 1–25.
- [6] Kang Wang, Adria Mamiidipalli, Tara Retson, Naeim Bahrami, Kyle Hasenstab, Kevin Blansit, Emily Bass, Timoteo Delgado, Guilherme Cunha, Michael S. Middleton, et al., Automated ct and mri liver segmentation and biometry using a generalized convolutional neural network, *Radiology: Artif. Intell.* 1 (2) (2019), 180022.
- [7] Mubashir Ahmad, Danni Ai, Guiwang Xie, Syed Furqan Qadri, Hong Song, Yong Huang, Yongtian Wang, Jian Yang, Deep belief network modeling for automatic liver segmentation, *IEEE Access* 7 (2019) 20585–20595.
- [8] Olaf Ronneberger, Philipp Fischer, Thomas Brox, U-net: convolutional networks for biomedical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2015, pp. 234–241.
- [9] Vijay Badrinarayanan, Alex Kendall, Roberto Cipolla, Segnet: a deep convolutional encoder-decoder architecture for image segmentation, *IEEE Trans. Pattern Anal. Mach. Intell.* 39 (12) (2017) 2481–2495.
- [10] Jeya Maria Jose Valanarasu, Vishwanath A. Sindagi, Ilker Hacıhaliloğlu, Vishal M. Patel, Kiu-net: towards accurate segmentation of biomedical images using over-complete representations, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2020, pp. 363–373.
- [11] Zgun iek, A. Abdulkadir, S.S. Lienkamp, T. Brox, O. Ronneberger, 3d U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation, Springer, Cham, 2016.
- [12] Cao Hu, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Tian Qi, Manning Wang, Swin-unet: Unet-like Pure Transformer for Medical Image Segmentation, 2021 arXiv preprint arXiv:2105.05537.
- [13] Wenxuan Wang, Chen Chen, Meng Ding, Yu Hong, Zha Sen, Jiangyun Li, Transbts: multimodal brain tumor segmentation using transformer, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2021, pp. 109–119.
- [14] Yuanfeng Ji, Ruimao Zhang, Huijie Wang, Zhen Li, Lingyun Wu, Shaoting Zhang, Ping Luo, Multi-compound transformer for accurate biomedical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2021, pp. 326–336.
- [15] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, L. Yuille Alan, Yuyin Zhou, Transunet: Transformers Make Strong Encoders for Medical Image Segmentation, 2021 arXiv preprint arXiv:2102.04306.
- [16] Jun Fu, Jing Liu, Haijie Tian, Yong Li, Yongjun Bao, Zhiwei Fang, Hanqing Lu, Dual attention network for scene segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 3146–3154.
- [17] A. Tsai, A. Yezzi, W. Wells, C. Tempany, D. Tucker, A. Fan, W.E. Grimson, A. Willsky, A shape-based approach to the segmentation of medical imagery using level sets, *IEEE Trans. Med. Imag.* 22 (2) (2003) 137, 137.
- [18] Moria Tau, Tal Hassner, Dense correspondences across scenes and scales, *IEEE Trans. Pattern Anal. Mach. Intell.* 38 (5) (2015) 875–888.
- [19] Tobias Heimann, Hans-Peter Meinzer, Ivo Wolf, A statistical deformable model for the segmentation of liver ct volumes, *3D Segment. Clin.: A grand challenge* (2007) 161–166.
- [20] Jonathan Long, Evan Shelhamer, Trevor Darrell, Fully convolutional networks for semantic segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 3431–3440.
- [21] Xiao Xiao, Lian Shen, Zhiming Luo, Shaozi Li, Weighted res-unet for high-quality retina vessel segmentation, in: *2018 9th International Conference on Information Technology in Medicine and Education (ITME)*, IEEE, 2018, pp. 327–331.
- [22] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, Jianming Liang, Unet++: a nested u-net architecture for medical image segmentation, in: *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, Springer, 2018, pp. 3–11.
- [23] Hengshuang Zhao, Xiaojuan Qi, Xiaoyong Shen, Jianping Shi, Ji-aya Jia, Icnets for real-time semantic segmentation on high-resolution images, in: *Proceedings of the European Conference on Computer Vision, ECCV*, 2018, pp. 405–420.
- [24] W. Chen, Z. Jiang, Z. Wang, K. Cui, X. Qian, Collaborative Global-Local Networks for Memory-Efficient Segmentation of Ultra-High Resolution Images, *IEEE*, 2019.
- [25] Liang-Chieh Chen, Papandreou George, Iasonas Kokkinos, Kevin Murphy, L. Yuille Alan, Semantic Image Segmentation with Deep Convolutional Nets and Fully Connected Crfs, 2014 arXiv preprint arXiv:1412.7062.
- [26] L.C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A. L. Yuille Deeplab, Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, *IEEE Trans. Pattern Anal. Mach. Intell.* 40 (4) (2018) 834–848.
- [27] Xiaolong Wang, Ross Girshick, Abhinav Gupta, Kaiming He, Non-local neural networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7794–7803.
- [28] S. Woo, J. Park, J.Y. Lee, I.S. Kweon, Cbam: Convolutional Block Attention Module, Springer, Cham, 2018.
- [29] Lei Mou, Yitian Zhao, Li Chen, Jun Cheng, Zaiwang Gu, Huaying Hao, Hong Qi, Yalin Zheng, Alejandro Frangi, Liu Jiang, Cs-net: channel and spatial attention network for curvilinear structure segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2019, pp. 721–730.
- [30] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [31] Sixiao Zheng, Jiachen Lu, Hengshuang Zhao, Xiatian Zhu, Zekun Luo, Yabiao Wang, Yanwei Fu, Jianfeng Feng, Tao Xiang, Philip H.S. Torr, et al., Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6881–6890.
- [32] Enze Xie, Wenjia Wang, Wenhai Wang, Peize Sun, Hang Xu, Liang Ding, Ping Luo, Segmenting Transparent Object in the Wild with Transformer, 2021 arXiv preprint arXiv:2101.08461.
- [33] Yuqing Wang, Zhaoliang Xu, Xinlong Wang, Chunhua Shen, Baoshan Cheng, Hao Shen, Huaxia Xia, End-to-end video instance segmentation with transformers, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 8741–8750.
- [34] Yuanxuan Li, Ziyu Wang, Li Yin, Zhiqin Zhu, Guanqiu Qi, Yu Liu, X-Net: a Dual Encoding-Decoding Method in Medical Image Segmentation, *The Visual Computer*, 2021, pp. 1–11.
- [35] Yang Xu, Xianyu He, Guofeng Xu, Guanqiu Qi, Yu Kun, Li Yin, Yang Pan, Yuehui Yin, Hao Chen, A medical image segmentation method based on multi-dimensional statistical features, *Front. Neurosci.* 16 (2022).
- [36] Zhiqin Zhu, Xianyu He, Guanqiu Qi, Yuanxuan Li, Baisan Cong, Yu Liu, Brain tumor segmentation based on the fusion of deep semantics and edge information in multimodal mri, *Inf. Fusion* 91 (2023) 376–387.
- [37] Hang Zhang, Chongruo Wu, Zhongyue Zhang, Yi Zhu, Haibin Lin, Zhi Zhang, Yue Sun, He Tong, Jonas Mueller, R. Manmatha, et al., Resnest: Split-Attention Networks, 2020 arXiv preprint arXiv:2004.08955.
- [38] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, 770778.
- [39] Yunhe Gao, Mu Zhou, Dimitris N. Metaxas, Unet: a hybrid transformer architecture for medical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2021, pp. 61–71.
- [40] Ze Liu, Yutong Lin, Yue Cao, Hu Han, Yixuan Wei, Zheng Zhang, Stephen Lin, Baining Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10012–10022.
- [41] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, Piotr Dollar, Focal loss for dense object detection, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [42] Leslie N. Smith, Cyclical Focal Loss, 2022 arXiv preprint arXiv:2202.08978.
- [43] M. Rema, R. Pradeepa, Diabetic retinopathy: an indian perspective, *Indian J. Med. Res.* 125 (2007) 297–310.
- [44] Lab Tao, A. Yg, Wac Kai, G.A. Song, D. Hl, Kae Hong, Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening - sciencedirect, *Inf. Sci.* 501 (2019) 511–522.
- [45] Noel CF. Codella, David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kallou, Konstantinos Liopyris, Nabin Mishra, Harald Kittler, et al., Skin lesion analysis toward melanoma detection: a challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic), in: *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, IEEE, 2018, pp. 168–172.
- [46] Paszke Adam, Sam Gross, Francisco Massa, Lerer Adam, Bradbury James, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al., Pytorch: an imperative style, high-performance deep learning library, *Adv. Neural Inf. Process. Syst.* 32 (2019).
- [47] L.C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with Atrous Separable Convolution for Semantic Image Segmentation, Springer, Cham, 2018.
- [48] S. Guo, T. Li, H. Kang, N. Li, Y. Zhang, K. Wang, L-seg: an end-to-end unified framework for multi-lesion segmentation of fundus images, *Neurocomputing* 349 (JUL.15) (2019) 52–63.
- [49] Azade Farshad, Yousef Yeganeh, Peter Gehlbach, Nassir Navab, Y-net: a spatiospectral dual-encoder network for medical image segmentation, in: *Medical Image Computing and Computer Assisted Intervention-MICCAI 2022: 25th International Conference, Singapore, September 18–22, 2022, Proceedings, Part II*, Springer, 2022, pp. 582–592.